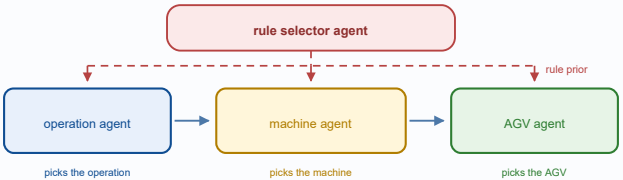
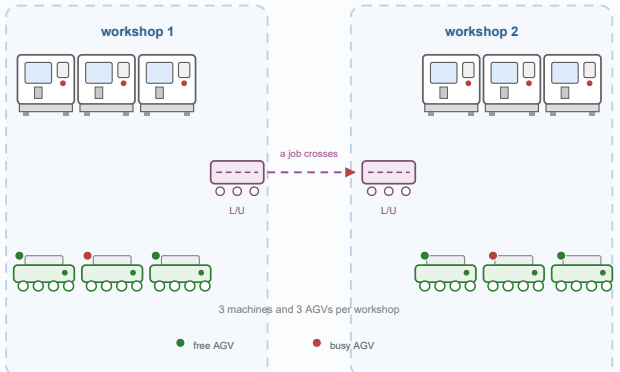
(a) Four agents, three coupled decisions

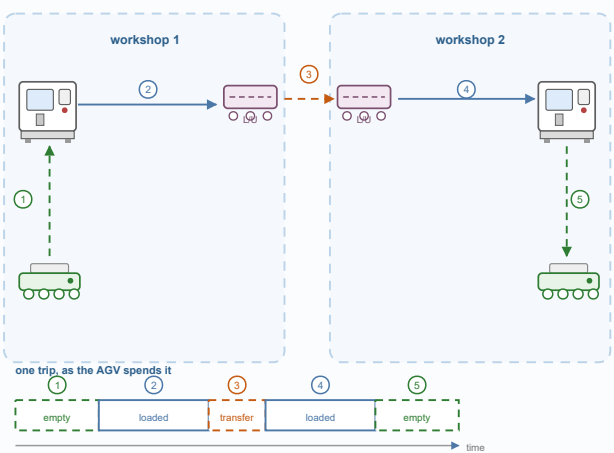


the weights set the trade-off; the transport term counts loaded travel plus half-weighted empty travel

(b) Shop environment

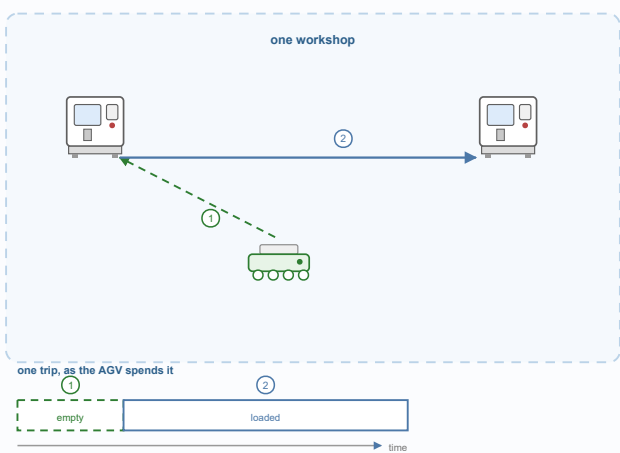


(c) Cross-workshop transport



the two loaded legs belong to different fleets; the transfer uses no AGV

(d) Same-workshop transport



one fleet, no station hop, so the trip has no transfer segment